PHYS1408 Practice Exam 3-1 — Answer Key

Worked solutions with course-material citations

Every numeric answer was computed and checked in `_scripts/build_practice-exam_3_1.py`. Constants are the ones printed on the 2015 Exam 3: $G = 6.67 \times 10^{-11}$, $M_{earth} = 5.98 \times 10^{24}$ kg, $R_{earth} = 6.37 \times 10^{6}$ m, and $g = 9.8$ m/s^2.

Sources. "Ch 12 notes p. 6/10" means page 6 of the professor's *Chapter 12 Statics* notes (he numbers his pages "6/10"). "2015 Exam 3 #2" is question 2 of the previous exam in `Professors-Notes/exam 3-4.pdf`.

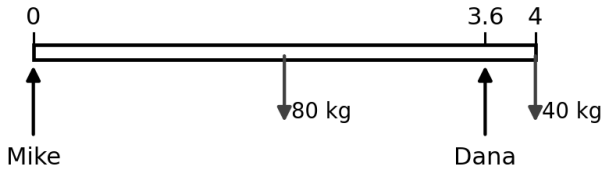
Practice Exam 3-1 follows the 2015 Exam 3 (its Ch 12 and 13 questions, #1–5 and #9–13) with new numbers, plus lecture-note examples: the diving board, the seesaw, the wall beam with a cable, and the orbit altitude from a period. Question 2 is a **maximum**: pick the largest choice that still works. Every question is a procedure worked in the notes or on the 2015 exam.

Q	Ans	Answer	Type	Topic (source)	Notecard
1	**d**	871 N	Calc	Two people carry a sofa (Ch 12 p. 4–5; 2015 #1)	Sofa / beam
2	**a**	5.0 m	Max	Man climbing a ladder, x_{max} (Ch 12 p. 8; 2015 #2)	Ladder
3	**e**	256 N	Calc	Light hung by two cords (2015 #3; Ch 12 p. 1)	Two cords
4	**b**	3.92 mm	Calc	Young's modulus, compressed block (2015 #4; Ch 12 p. 10)	Moduli
5	**e**	0.0250 m	Calc	Young's modulus, wire stretch (Ch 12 p. 10; 2015 #5)	Moduli
6	**c**	15,700 N	Calc	Diving board, support B (Ch 12 p. 3)	Diving board
7	**b**	0.625 m	Calc	Seesaw balance (Ch 12 p. 4)	Seesaw

Q	Ans	Answer	Type	Topic (source)	Notecard
8	**d**	2.58 m	Calc	Beam on a wall hinge with a cable (x_{max}) (Ch 12 p. 6)	Wall beam + cable
9	**a**	8.45 m/s^2	Calc	g at an altitude (Ch 13 p. 1–2; 2015 #12)	Gravity
10	**e**	11.8 hours	Calc	Satellite period (Ch 13 p. 2; 2015 #10)	Orbit
11	**c**	6.04 hours	Calc	Binary stars, period (Ch 13 p. 4; 2015 #11)	Binary stars
12	**c**	3.28 km/s	Calc	Launch speed to reach a height (Ch 13 p. 7–8; 2015 #9)	Rocket
13	**d**	89.6%	Calc	Equal-pull point (2015 #13; Ch 13 p. 1)	Equal pull
14	**b**	2.38 km/s	Calc	Escape speed (Ch 13 p. 9)	Escape
15	**a**	13,900 km	Calc	Orbit altitude from the period (Ch 13 p. 3)	Orbit

1. Two people carry a sofa — d) 871 N

Source: Ch 12 notes p. 4/10 (two men carry a sofa: $\sum \tau_{Jim}$, $\sum \tau_{Sam}$) and p. 5/10 (20 kg kid set on the sofa 1 m from Jim).
Previous-exam match: 2015 Exam 3 #1 (sofa + box of books on Sarah's end), new numbers
Notecard: Sofa / beam
What it tests: torque balance about one carrier to find the other carrier's force.



Step 1: pick the pivot at Mike's hands. A force applied at the pivot has $r = 0$, so its torque is zero and it drops out of the equation. The notes do this on p. 3/10 ("A is eliminated").

Step 2: torque balance about Mike. Every force here is vertical and the sofa is horizontal, so each torque is $\tau = rF$, with r measured from Mike. The sofa's weight acts at its center, $L/2$. Dana's upward push is counterclockwise; the weights are clockwise.

$$\sum \tau_{Mike} = 0: \quad x_{Dana}\, F_{Dana} = \tfrac{L}{2}\, Mg + x_1\, m_1 g$$

Step 3: plug in ($x_{Dana} = 3.6$ m):

$$3.6 \text{ m} \times F_{Dana} = \underbrace{2 \text{ m}}_{\text{sofa center}}\ \underbrace{(80)(9.8)}_{\text{sofa weight}} + \underbrace{4 \text{ m}}_{\text{box of books position}}\ \underbrace{(40)(9.8)}_{\text{box of books weight}}$$

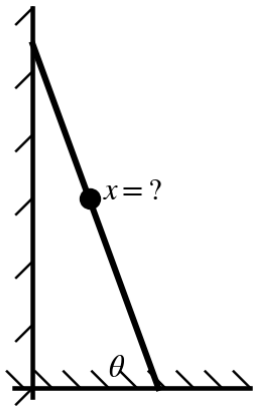
$$F_{Dana} = \frac{3{,}136 \text{ N m}}{3.6 \text{ m}} = 871.1 \text{ N}$$

Why the other choices are wrong:

- a) 305 N: is the force on Mike, the other person.
- b) 436 N: leaves out the sofa's own weight.
- c) 588 N: splits the total weight evenly; that is only true with nothing extra and both at the ends.
- e) 1,180 N: is the total weight of everything; the two people share it.

2. Man climbing a ladder, x_{max} — a) 5.0 m

Source: Ch 12 notes p. 8/10 (leaning ladder with a man climbing it, $x = \frac{L}{M}(\mu_s(M+m)\tan\theta - \frac{m}{2}) = 8.8$ m).
Previous-exam match: 2015 Exam 3 #2 (150 kg man, 20 kg ladder, 71°, $\mu = 0.25$, 5 m), new numbers
Notecard: Ladder
What it tests: ladder torque balance with a climber, then the maximum-answer rule.



Step 1: forces (notes p. 8/10). $\sum F_y$: $N = (M+m)g$. $\sum F_x$: $N_w = F_f$, and at the limit $F_f = \mu_s N = \mu_s(M+m)g$.

Step 2: torque about the base. Ladder weight at $L/2$ (arm $\frac{L}{2}\cos\theta$), man at x (arm $x\cos\theta$), wall force at the top (arm $L\sin\theta$):

$$\tfrac{L}{2}\cos\theta\, mg + x\cos\theta\, Mg - L\sin\theta\,\mu_s(M+m)g = 0$$

Cancel g, divide by $\cos\theta$:

$$x = \frac{L}{M}\Big(\mu_s(M+m)\tan\theta - \frac{m}{2}\Big)$$

Step 3: plug in ($\tan 70^\circ = 2.747$):

$$x = \frac{\overbrace{6}^{L}}{\underbrace{120}_{M}}\Big(\underbrace{(0.3)}_{\mu_s}\,\underbrace{(145)}_{M+m}(2.747) - \underbrace{\tfrac{25}{2}}_{m/2}\Big) = 5.35 \text{ m}$$

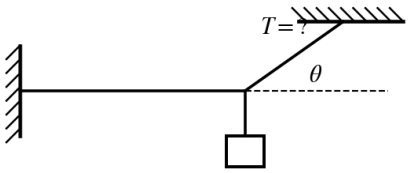
Step 4: it's a maximum. He can go up to 5.35 m; any farther and the ladder slips. The largest choice that still works is **5.0 m**. (On the 2015 exam, $x = 3.78$ m and the answer was b) 3.5 m.)

Why the other choices are wrong:

- b) 5.5 m: is more than the maximum, so it fails.
- c) 6.0 m: is more than the maximum, so it fails.
- d) 6.5 m: is more than the maximum, so it fails.
- e) 7.0 m: is more than the maximum, so it fails.

3. Light hung by two cords — e) 256 N

Source: 2015 Exam 3 #3, solved with $\sum F_x = \sum F_y = 0$ from Ch 12 notes p. 1/10 ("Static means not moving").
Previous-exam match: 2015 Exam 3 #3 (20 kg light, 40° with the ceiling: 305 N → b) 300 N), new numbers
Notecard: Two cords
What it tests: $\sum F_y = 0$ on the knot: $T \sin\theta = mg$.



Step 1: forces on the knot (static, so $\sum F_x = \sum F_y = 0$, notes p. 1/10). Ceiling cord T at θ above the horizontal; horizontal cord T_h; the light's weight mg hangs straight down.

Step 2: $\sum F_y = 0$. Only the ceiling cord has a vertical part, $T \sin\theta$:

$$T \sin\theta = mg \;\Rightarrow\; T = \frac{mg}{\sin\theta} = \frac{\overbrace{(15)(9.8)}^{mg=147.0\text{ N}}}{\sin 35^\circ} = 256.3 \text{ N}$$

Why the other choices are wrong:

- a) 84.3 N: multiplies by $\sin\theta$ instead of dividing.
- b) 147 N: is just the light's weight.
- c) 179 N: uses $\cos\theta$: that would be the angle from the vertical.
- d) 210 N: is the tension in the horizontal cord.

4. Young's modulus, compressed block — b) 3.92 mm

Source: 2015 Exam 3 #4, solved with Young's modulus from Ch 12 notes p. 10/10.
Previous-exam match: 2015 Exam 3 #4 (jello 3 m × 4 m × 0.1 m, 500 kg: 1.36 mm → e) 1.3 mm), new numbers
Notecard: Moduli
What it tests: compressive stress on the top face, strain = $\Delta h/h$.

Step 1: the formula (Ch 12 notes p. 10/10), with compression instead of stretch:

$$Y = \frac{F/A}{\Delta h/h} \;\Rightarrow\; \Delta h = \frac{F\,h}{A\,Y}$$

Step 2: the pieces. The mass pushes down on the **top** face, so A = width × length. h is the original height.

$$F = (400)(9.8) = 3{,}920 \text{ N}$$

$$A = (2)(3) = 6.000 \text{ m}^2$$

Step 3: plug in:

$$\Delta h = \frac{\underbrace{(3{,}920)}_{F}\,\underbrace{(0.15)}_{h}}{\underbrace{(6.000)}_{\text{top area}}\,\underbrace{(25{,}000)}_{Y}} = 0.00392 \text{ m} = 3.92 \text{ mm}$$

Why the other choices are wrong:

- a) 0.400 mm: forgets g.
- c) 26.1 mm: finds the strain and stops (forgets to multiply by the height).
- d) 52.3 mm: uses the other side face ($l \times h$).
- e) 78.4 mm: uses a side face ($w \times h$) as the area.

5. Young's modulus, wire stretch — e) 0.0250 m

Source: Ch 12 notes p. 10/10 (Young's modulus $Y = \frac{F/A}{\Delta\ell/\ell}$).
Previous-exam match: 2015 Exam 3 #5 (500 kg, 4.00 m, 0.001 m diameter, 8×10^{10}: 0.312 m $\rightarrow$ e) 0.3 m), new numbers
Notecard: Moduli
What it tests: tensile stress/strain with a circular cross-section ($r = d/2$).

Step 1: the formula (Ch 12 notes p. 10/10):

$$Y = \frac{\text{tensile stress}}{\text{tensile strain}} = \frac{F/A}{\Delta\ell/\ell}$$

Step 2: the pieces. The force is the load's weight, and the area is the wire's circular cross-section (radius = half the diameter):

$$F = mg = (300)(9.8) = 2{,}940 \text{ N}$$

$$A = \pi r^2 = \pi\Big(\underbrace{\tfrac{0.0015}{2}}_{r}\Big)^2 = 1.767 \times 10^{-6} \text{ m}^2$$

Step 3: solve for the stretch:

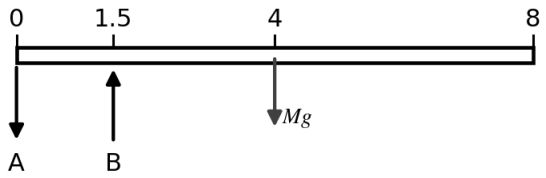
$$\Delta\ell = \frac{F\,\ell}{A\,Y} = \frac{\underbrace{(2{,}940)}_{F}\,\underbrace{(3)}_{\ell}}{\underbrace{(1.767 \times 10^{-6})}_{A}\,\underbrace{(2 \times 10^{11})}_{Y}} = 0.0250 \text{ m}$$

Why the other choices are wrong:

- a) 0.00255 m: forgets g (uses the mass instead of the weight).
- b) 0.00624 m: uses the diameter as the radius in $A = \pi r^2$.
- c) 0.00832 m: finds the strain $\Delta\ell/\ell$ and stops.
- d) 0.0196 m: uses $A = d^2$ (no π, no half).

6. Diving board, support B — c) 15,700 N

Source: Ch 12 notes p. 3/10 (diving board, $L = 10$ m, $M = 1000$ kg: $B = 5Mg$, $A = 4Mg$).
Previous-exam match: none (lecture-notes example)
Notecard: Diving board
What it tests: choosing the pivot to eliminate one support, then $\sum F_y = 0$ (A pulls down).



Step 1: draw the forces (notes p. 3/10). The board's weight Mg acts at its center, $L/2 = 4$ m. B pushes **up**. A must pull **down**, or the board would tip off B.

Step 2: torque about A removes A ($r = 0$):

$$\sum \tau_A = 0: \quad a\,B - \tfrac{L}{2}\,Mg = 0$$

$$B = \frac{L/2}{a}\,Mg = \frac{\overbrace{4\text{ m}}^{L/2}}{\underbrace{1.5\text{ m}}_{\text{A to B}}}\,\underbrace{(600)(9.8)}_{Mg=5{,}880\text{ N}} = 15{,}680\text{ N}$$

Step 3: $\sum F_y = 0$ with B up and A, Mg down:

$$B - A - Mg = 0 \;\Rightarrow\; A = B - Mg = 15{,}680 - 5{,}880 = 9{,}800\text{ N}$$

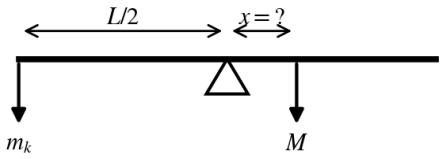
In the notes ($L = 10$ m, $a = 1$ m) this gives $B = 5Mg$ and $A = 4Mg$.

Why the other choices are wrong:

- a) 5,880 N: is just the board's weight Mg.
- b) 9,800 N: is the force from support A, not B.
- d) 21,600 N: adds Mg instead of subtracting it.
- e) 31,400 N: puts the board's weight at the far end (L) instead of its center ($L/2$).

7. Seesaw balance — b) 0.625 m

Source: Ch 12 notes p. 4/10 (seesaw, $m_k = 20$ kg, $M = 100$ kg, $x = L/10$).
Previous-exam match: none (lecture-notes example)
Notecard: Seesaw
What it tests: torque balance about the pivot; the seesaw's own mass drops out.



Step 1: torque about the center (pivot). The seesaw's own weight acts at the center, so its lever arm is zero. That's why the notes say its mass "don't care!" The kid is $L/2$ from the pivot; the adult is x from it.

$$\sum \tau_{center} = 0: \quad \tfrac{L}{2}\, m_k g - x\, M_T g = 0 \;\Rightarrow\; x = \frac{L\, m_k}{2 M_T}$$

g cancels.

Step 2: plug in:

$$x = \frac{\underbrace{5\text{ m}}_{\text{length}}\ \underbrace{20\text{ kg}}_{\text{kid}}}{2\ \underbrace{80\text{ kg}}_{\text{adult}}} = 0.625\text{ m}$$

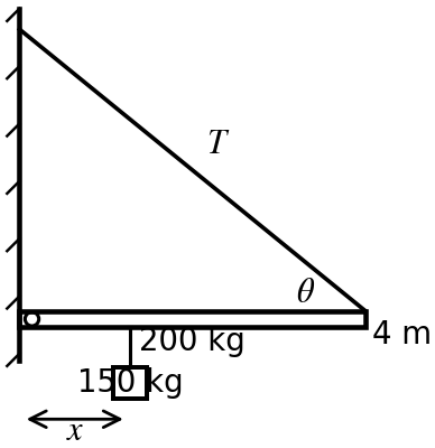
In the notes, $m_k = 20$ kg and $M_T = 100$ kg give $x = L/10$.

Why the other choices are wrong:

- a) 0.417 m: adds the seesaw's mass to the adult's; its weight acts at the pivot and has no torque.
- c) 1.25 m: forgets that the kid's lever arm is $L/2$, not L.
- d) 1.88 m: measures from the end of the seesaw instead of from the center.
- e) 3.12 m: adds the half-length.

8. Beam on a wall hinge with a cable (x_{max}) — d) 2.58 m

Source: Ch 12 notes p. 6/10 (beam hinged at the wall, cable at $\theta = 30°$, find x_{max}); $\tau = rF_{\perp}$ from p. 1/10.
Previous-exam match: none (lecture-notes example)
Notecard: Wall beam + cable
What it tests: torque about the hinge with the perpendicular part of the tension.



Step 1: torque about the hinge (left end). The notes say "Do torque at left to remove reaction forces": the hinge forces R_x, R_y act at $r = 0$.

Only the **perpendicular** part of the tension, $T\sin\theta$, makes torque at the far end ($\tau = rF_{\perp}$, notes p. 1/10). The beam's weight acts at $L/2$; the mass's weight acts at x.

$$\sum \tau_{left} = 0: \quad -x\,mg - \tfrac{L}{2}\,Mg + L\,T\sin\theta = 0$$

Step 2: solve for x:

$$x = \frac{L}{mg}\left(T\sin\theta - \frac{Mg}{2}\right)$$

Step 3: plug in $T = T_{max}$:

$$x_{max} = \underbrace{\frac{4\text{ m}}{(150)(9.8)}}_{mg}\left(\underbrace{(3{,}000)(\sin 40°)}_{T\sin\theta = 1{,}928\text{ N}} - \underbrace{\frac{(200)(9.8)}{2}}_{Mg/2 = 980.0\text{ N}}\right) = 2.58\text{ m}$$

Notes numbers: $L = 5$ m, $T_{max} = 10^4$ N, $M = 1000$ kg, $m = 100$ kg, $\theta = 30°$ give $x_{max} = 0.510$ m.

Why the other choices are wrong:

- a) 0.658 m: No standard mistake gives this value; it is a spacing choice.
- b) 0.921 m: No standard mistake gives this value; it is a spacing choice.
- c) 1.29 m: divides by 2 one time too many.
- e) 3.59 m: uses $\cos\theta$ instead of $\sin\theta$ for the cable's perpendicular part.

9. g at an altitude — a) 8.45 m/s²

Source: Ch 13 notes p. 1/10 ($g = GM_E/R_E^2 = 9.81$ m/s²) and p. 2/10 ($r = R + h$).
Previous-exam match: 2015 Exam 3 #12 (750 km: 7.87 m/s² → b) 8 m/s²), new numbers
Notecard: Gravity
What it tests: $g = GM/r^2$ with r measured from Earth's center.

Step 1: the formula (notes p. 1/10): $mg = \dfrac{GMm}{R^2}$, so $g = \dfrac{GM}{R^2}$ at the surface. Above the surface the distance is measured from Earth's **center**: $r = R + h$ (notes p. 2/10).

$$g = \frac{GM_{earth}}{(R+h)^2}$$

Step 2: distance from the center:

$$r = \underbrace{6.37 \times 10^6}_{R_{earth}} + \underbrace{5 \times 10^5}_{h \text{ in m}} = 6.870 \times 10^6 \text{ m}$$

Step 3: plug in:

$$g = \frac{\underbrace{(6.67 \times 10^{-11})}_{G}\,\underbrace{(5.98 \times 10^{24})}_{M_{earth}}}{(6.870 \times 10^6)^2} = 8.45 \text{ m/s}^2$$

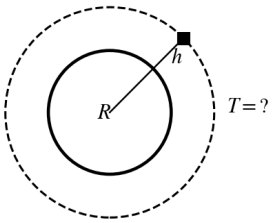
The person's mass cancels ($mg = GMm/r^2$); it's extra information.

Why the other choices are wrong:

- b) 9.83 m/s²: uses R_{earth} (surface g), ignoring the height.
- c) 1,600 m/s²: uses h instead of $R + h$ (the 2015 exam's e) 700 m/s² is this mistake).
- d) 2,240 m/s²: No standard mistake gives this value; it is a spacing choice.
- e) 3,140 m/s²: No standard mistake gives this value; it is a spacing choice.

10. Satellite period — e) 11.8 hours

Source: Ch 13 notes p. 2/10 (satellite orbiting a planet, $T^2 = 4\pi^2 r^3/GM$, $r = R + h$; $h = 400$ km $\to$ 92.4 min).
Previous-exam match: 2015 Exam 3 #10 (altitude 10^8 m: 3.99 days $\to$ d) 4 days), new numbers
Notecard: Orbit
What it tests: $T^2 = 4\pi^2 r^3/GM$ with $r = R + h$.



Step 1: gravity supplies the centripetal force (notes p. 2/10):

$$\frac{GMm}{r^2} = \frac{mv^2}{r}, \qquad v = \frac{2\pi r}{T} \;\Rightarrow\; T^2 = \frac{4\pi^2 r^3}{GM}$$

The satellite's mass m cancels.

Step 2: orbit radius from Earth's center:

$$r = \underbrace{6.37 \times 10^6}_{R} + \underbrace{2 \times 10^7}_{h} = 2.637 \times 10^7 \text{ m}$$

Step 3: plug in:

$$T = \sqrt{\frac{4\pi^2 (2.637 \times 10^7)^3}{\underbrace{(6.67 \times 10^{-11})(5.98 \times 10^{24})}_{GM_{earth}}}} = 42{,}600 \text{ s}$$

Step 4: convert: 42,600 s = 11.8 hours.

Notes example: $h = 400$ km gives $T = 5546$ s = 92.4 min (low Earth orbit).

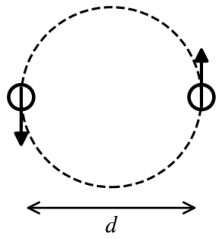
Why the other choices are wrong:

- a) 1.40 hours: uses R_{earth} (an orbit skimming the surface).
- b) 1.88 hours: drops the $4\pi^2$ (forgets the 2π in $v = 2\pi r/T$).

- c) 5.92 hours: uses π instead of 2π.
- d) 7.82 hours: uses h instead of $r = R + h$.

11. Binary stars, period — c) 6.04 hours

Source: Ch 13 notes p. 4/10 (two equal-mass stars, $T^2 = 16\pi^2 r^3/GM$; "separation btwn the masses is $2r$ (not r)").
Previous-exam match: 2015 Exam 3 #11 (10^{31} kg each, $d = 10^8$ m: 2.9 min $\rightarrow$ a) 3 minutes), new numbers
Notecard: Binary stars
What it tests: $T^2 = 16\pi^2 r^3/GM$ with r = half the separation.



Step 1: the force uses the separation, the circle uses r (notes p. 4/10). Each star moves on a circle of radius r about the center; the distance between the stars is $2r$:

$$\frac{GM^2}{(2r)^2} = \frac{Mv^2}{r}, \qquad v = \frac{2\pi r}{T} \;\Rightarrow\; T^2 = \frac{16\pi^2 r^3}{GM}$$

Step 2: the orbit radius: $r = d/2 = 1 \times 10^9$ m.

Step 3: plug in:

$$T = \sqrt{\frac{16\pi^2 (1\times 10^9)^3}{\underbrace{(6.67\times 10^{-11})}_{G}\,\underbrace{(5\times 10^{30})}_{M}}} = 21{,}760 \text{ s} = 6.04 \text{ hours}$$

Notes example: $M = 2 \times 10^{30}$ kg, $r = 6 \times 10^{10}$ m gives 185 days.

Why the other choices are wrong:

- a) 3.02 hours: uses the single-planet $4\pi^2$ formula (forgets that the separation is $2r$ in the force).
- b) 4.27 hours: uses $2M$ for the mass.

- d) 8.55 hours: uses the single-planet formula with $r = d$.
- e) 17.1 hours: uses the separation d as the orbit radius (the notes warn: separation is $2r$, not r).

12. Launch speed to reach a height — c) 3.28 km/s

Source: Ch 13 notes p. 7/10 ($U = -GMm/r$, energy conservation) and p. 8/10 ("given a target altitude $h = 1000$ km, what v to launch?" 4119 m/s).
Previous-exam match: 2015 Exam 3 #9 (1000 km), new numbers
Notecard: Rocket
What it tests: energy conservation with $U = -GMm/r$, solved for v.

Step 1: conservation of energy (notes p. 7–8/10). At launch $r = R$, speed v; at the top $r = R + h$, speed 0. Gravitational PE is $U = -GMm/r$:

$$-\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{r} \;\Rightarrow\; v = \sqrt{2GM\Big(\frac{1}{R} - \frac{1}{r}\Big)}$$

Step 2: top of the flight: $r = R + h = 6.37 \times 10^6 + 6 \times 10^5 = 6.970 \times 10^6$ m.

Step 3: plug in:

$$v = \sqrt{2\,\underbrace{(6.67 \times 10^{-11})(5.98 \times 10^{24})}_{GM_{earth}}\Big(\underbrace{\frac{1}{6.37 \times 10^6}}_{1/R} - \underbrace{\frac{1}{6.970 \times 10^6}}_{1/r}\Big)} = 3{,}283 \text{ m/s} = 3.28 \text{ km/s}$$

Notes example: $h = 1000$ km needs $v = 4119$ m/s.

Why the other choices are wrong:

- a) 1.66 km/s: No standard mistake gives this value; it is a spacing choice.
- b) 2.32 km/s: forgets the 2 from $\frac{1}{2}mv^2$.
- d) 7.56 km/s: is the orbital speed at that height, not the launch speed.
- e) 11.2 km/s: is the escape speed.

13. Equal-pull point — d) 89.6%

Source: 2015 Exam 3 #13, solved with $F = GMm/r^2$ from Ch 13 notes p. 1/10.
Previous-exam match: 2015 Exam 3 #13 (moon 7×10^{22} kg, 400,000 km: 90.2% → b) 90%), new numbers
Notecard: Equal pull
What it tests: setting two inverse-square pulls equal: $x/d = k/(1+k)$, $k = \sqrt{M_E/M_b}$.

Step 1: set the two pulls equal with Newton's law (Ch 13 notes p. 1/10). Let x be the distance from Earth's center, so the distance to the moon is $d - x$. The test mass m cancels:

$$\frac{GM_E m}{x^2} = \frac{GM_b m}{(d-x)^2} \;\Rightarrow\; \frac{x}{d-x} = \sqrt{\frac{M_E}{M_b}}$$

Step 2: the square root:

$$k = \sqrt{\frac{\overbrace{5.98 \times 10^{24}}^{M_E}}{\underbrace{8 \times 10^{22}}_{M_b}}} = 8.646$$

Step 3: solve for the fraction: $x = k(d-x)$, so

$$\frac{x}{d} = \frac{k}{1+k} = \frac{8.646}{9.646} = 0.8963 = 89.6\%$$

The distance doesn't matter for the percentage; only the mass ratio does.

Why the other choices are wrong:

- a) 10.4%: gives the distance from the moon's center.
- b) 50.0%: assumes the halfway point.
- c) 80.8%: uses a cube root.
- e) 98.7%: uses the mass ratio without the square root.

14. Escape speed — b) 2.38 km/s

Source: Ch 13 notes p. 9/10 ($v_{escape} = \sqrt{2GM/R} = 11.2$ km/s for Earth).
Previous-exam match: none (lecture-notes example)
Notecard: Escape
What it tests: $v_{esc} = \sqrt{2GM/R}$.

Step 1: energy conservation (notes p. 9/10). To escape, the object just reaches $r = \infty$ with $v = 0$, where $U = 0$ and $K = 0$:

$$-\frac{GMm}{R} + \frac{mv^2}{2} = 0 + 0 \;\Rightarrow\; v_{esc} = \sqrt{\frac{2GM}{R}}$$

Step 2: plug in:

$$v_{esc} = \sqrt{\frac{2\,\underbrace{(6.67\times10^{-11})}_{G}\,\underbrace{(7.4\times10^{22})}_{M}}{\underbrace{1.74\times10^{6}}_{R}}} = 2{,}382 \text{ m/s} = 2.38 \text{ km/s}$$

For Earth the notes get 11,181 m/s, about 11.2 km/s.

Why the other choices are wrong:

- a) 1.68 km/s: forgets the 2 (that is the orbital speed at the surface).
- c) 4.76 km/s: doubles the answer.
- d) 6.66 km/s: No standard mistake gives this value; it is a spacing choice.
- e) 9.32 km/s: No standard mistake gives this value; it is a spacing choice.

15. Orbit altitude from the period — a) 13,900 km

Source: Ch 13 notes p. 3/10 (geosynchronous orbit: $T = 24$ h $\rightarrow r = 4.22\times10^7$ m, $h = 35{,}900$ km).
Previous-exam match: none (lecture-notes example)
Notecard: Orbit
What it tests: the geosynchronous-orbit procedure (notes p. 3): $r = (T^2GM/4\pi^2)^{1/3}$, then $h = r - R$.

Step 1: solve the period formula for r (notes p. 3/10, geosynchronous orbit):

$$T^2 = \frac{4\pi^2 r^3}{GM} \;\Rightarrow\; r = \left(\frac{T^2GM}{4\pi^2}\right)^{1/3}$$

Step 2: period in seconds: $T = 8\text{ h} \times 3600\text{ s/h} = 28{,}800\text{ s}$.

Step 3: plug in:

$$r = \left(\frac{\underbrace{(28{,}800)^2}_{T^2}\ \underbrace{(6.67 \times 10^{-11})(5.98 \times 10^{24})}_{GM_{earth}}}{4\pi^2} \right)^{1/3} = 2.031 \times 10^7\text{ m}$$

Step 4: altitude above the surface:

$$h = r - R = 2.031 \times 10^7 - 6.37 \times 10^6 = 1.39 \times 10^7\text{ m} = 13{,}900\text{ km}$$

Notes example: $T = 24$ h gives $r = 4.22 \times 10^7$ m and $h = 35{,}900$ km.

Why the other choices are wrong:

- b) 20,300 km: gives r (from Earth's center); forgets to subtract R.
- c) 62,800 km: drops the $4\pi^2$.
- d) 91,500 km: takes a square root instead of a cube root.
- e) 1.28×10^5 km: No standard mistake gives this value; it is a spacing choice.